

CHEMISTRY CH.4 — CARBON AND ITS COMPOUNDS — FULL MIND MAP

Topic-wise · Fully Explained · Hinglish · SSC / BPSC / BSSC / Railway

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1. Carbon: Electronic Configuration & Basics
2. Carbon Allotropes: Diamond, Graphite, Fullerene & Graphene
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1. Carbon: Electronic Configuration & Basics

- Carbon ka atomic number 6; electronic configuration: $1s^2 2s^2 2p^2$; simplified 2, 4. [Q448, Q450]
- Carbon ke paas 2 shells hain (K aur L); valence electrons = 4; isliye tetravalent hai. [Q448]
- Carbon non-metal hai (Gold, Platinum, Copper ke opposite jo metals hain — carbon inme se belong nahi karta); human body ka 18% mass; high melting point. [Q448, Q480, Q506]
- Uses: diamonds (jewelry), inks, paints, carbon dating. [Q448]
- Carbon 4 covalent bonds bana sakta hai; huge variety of organic compounds possible — carbon compounds mein hamesha covalent bonding hoti hai, ionic bond generally nahi paayi jaati. [Q450, Q452, Q476, Q466]
- Carbon dioxide (CO_2) mein carbon ki valency 4 hoti hai. [Q508]
- Carbon apna nucleus chhota hone ki wajah se shared electron pairs ko strongly hold karta hai — isiliye strong bonds banata hai. [Q482, Q498]
- Catenation = carbon ki property jiski wajah se uske atoms aapas mein jud kar lambi chains bana sakte hain. [Q510]
- Catenation property aur tetravalency dono milkar carbon compounds ki bahut badi variety possible banate hain. [Q471]
- Silicon (Si) carbon jaisi catenation property dikhata hai (thodi kam extent mein). [Q462]
- Organic carbon compounds ka melting point ionic compounds se kam hota hai kyunki inke molecules ke beech intermolecular forces weak hoti hain (covalent molecular structure). [Q464]
- Do ya zyada elements combine hoke ek compound banate hain. [Q514]

2. Carbon Allotropes: Diamond, Graphite, Fullerene & Graphene

- Allotropes = same element, different structural forms; different physical properties. [Q451, Q453, Q455]
- Diamond: Har carbon sp^3 hybridized; tetrahedral 3D network; har carbon atom 4 other carbon atoms se bonded hota hai. Hardest natural substance; bad conductor (no free electrons). Jewelry, cutting tools, glass cutting aur

rock drilling mein use. [Q451, Q453, Q492, Q502, Q512]

- Diamond ko synthetically bhi bana sakte hain — pure carbon ko very high pressure aur high temperature dekar. [Q484]
- Graphite: Har carbon sp^2 hybridized; hexagonal layers — har carbon atom same plane mein 3 other carbon atoms se bonded hota hai. Layers ke beech weak van der Waals forces; slippery. Good conductor (delocalized π electrons); pencil lead, lubricant, electrodes. [Q453, Q455, Q504]
- Fullerene (C_{60}): Soccer ball/football jaisi cage-like shape; 12 pentagons + 20 hexagons. Medical applications, nanotechnology, superconductors. [Q455, Q457, Q478]
- Graphene = single layer of graphite; excellent conductor, strong, flexible. [Q457]
- Aerographene duniya ka sabse halka (lightest) material hai — carbon ka ek advanced nano-structure. [Q496]
- Coal carbon ka allotrope nahi hai — Diamond, Graphite, Fullerene, Graphene sab allotropes hain lekin Coal ek mixture hai (allotrope nahi). [Q500]
- Coke (carbon ka ek form) metallurgy mein reducing agent ke roop mein use hota hai (jaise blast furnace mein iron extraction). [Q516]
- ⚠ Trap: Diamond aur graphite donon carbon ke allotropes hain lekin properties bilkul alag — structure ki wajah se. [Q451, Q453]

3. Organic Chemistry Basics: Hydrocarbons

- Hydrocarbons = sirf carbon aur hydrogen se bane compounds. [Q452, Q454]
- Saturated hydrocarbons (Alkanes): C-C single bonds; general formula C_nH_{2n+2} . Methane CH_4 , Ethane C_2H_6 , Propane C_3H_8 , Butane C_4H_{10} . [Q454, Q456]
- Unsaturated hydrocarbons: Alkenes — C=C double bond; formula C_nH_{2n} — Ethene C_2H_4 , Propene C_3H_6 . Alkynes — $C\equiv C$ triple bond; formula C_nH_{2n-2} — Ethyne C_2H_2 , Propyne C_3H_4 . [Q456, Q458]
- Cycloalkanes: ring structure; saturated; formula C_nH_{2n} . Cyclohexane C_6H_{12} : 6 C-C bonds + 12 C-H bonds = 18 total covalent bonds. [Q449]
- Aromatic hydrocarbons: benzene ring (C_6H_6); alternate double bonds; resonance stabilized. [Q458, Q460]
- Sucrose molecule mein 12 carbon atoms hote hain ($C_{12}H_{22}O_{11}$). [Q490]

4. Functional Groups & Homologous Series

- Functional group = atom/group jo molecule ki chemical properties decide karta hai. [Q459, Q461, Q463]
- Common functional groups: Alcohol ($-OH$) — ethanol C_2H_5OH , suffix -ol. Aldehyde ($-CHO$) — methanal $HCHO$, ethanal CH_3CHO , suffix -al (jaise Pentanal — 5-carbon aldehyde). Ketone ($>C=O$) — propanone CH_3COCH_3 , suffix -one. Carboxylic acid ($-COOH$) — ethanoic acid CH_3COOH , suffix -oic acid. Halide ($-X$) — chloromethane CH_3Cl , prefix chloro-. Amine ($-NH_2$) — methanamine CH_3NH_2 , suffix -amine. [Q461, Q463, Q465, Q474]
- Homologous series = same functional group; successive members mein CH_2 ka difference. Alkanes: CH_4 , C_2H_6 , C_3H_8 ... Alcohols: CH_3OH , C_2H_5OH , C_3H_7OH ... [Q465, Q467]

5. IUPAC Nomenclature & Common Names

- Naming: prefix (carbon count) + root word (ane/ene/yne) + suffix (functional group). [Q467]
- Methane = CH_4 ; Ethane = C_2H_6 ; Propane = C_3H_8 ; Butane = C_4H_{10} . [Q454]
- Ethene = C_2H_4 ; Propene = C_3H_6 ; Ethyne = C_2H_2 ; Propyne = C_3H_4 . [Q456, Q458]
- Methanol = CH_3OH ; Ethanol = C_2H_5OH . [Q461]

- Methanal = HCHO ; Ethanal = CH_3CHO . [Q463]
- Propanone = CH_3COCH_3 ; Butanone = $\text{CH}_3\text{COC}_2\text{H}_5$. [Q465]
- Methanoic acid = HCOOH ; Ethanoic acid = CH_3COOH . [Q467]
- Common names: Ethanol = ethyl alcohol; Ethanoic acid = acetic acid; Propanone = acetone. [Q467, Q469]

6. Isomerism in Carbon Compounds

- Isomers = same molecular formula, different structural formula/arrangement. [Q468, Q470, Q472]
- Structural isomerism: Chain isomerism — different carbon skeleton (Butane vs Isobutane). Position isomerism — same functional group, different position (Propan-1-ol vs Propan-2-ol). Functional group isomerism — different functional groups (Ethanol vs Dimethyl ether). [Q470, Q472]
- Butane (C_4H_{10}) ke 2 isomers: n-butane aur isobutane. [Q472]
- Pentane (C_5H_{12}) ke 3 isomers. [Q472]
- ⚠ Isomers ki molecular formula same hoti hai lekin properties alag ho sakti hain. [Q468, Q470]

7. Chemical Properties: Combustion, Oxidation, Addition & Substitution

- Combustion: Carbon compounds oxygen mein burn karke $\text{CO}_2 + \text{H}_2\text{O} + \text{Heat}$ dete hain. $\text{CH}_4 + 2\text{O}_2 \rightarrow \text{CO}_2 + 2\text{H}_2\text{O}$. Saturated = clean flame; Unsaturated = sooty flame (kam hydrogen). [Q473, Q475]
- Carbon apne saare allotropic forms mein oxygen mein jalke Carbon dioxide (CO_2) deta hai. [Q494]
- Oxidation: Alcohols $\rightarrow$ Aldehydes $\rightarrow$ Carboxylic acids. $\text{C}_2\text{H}_5\text{OH} + [\text{O}] \rightarrow \text{CH}_3\text{CHO} + \text{H}_2\text{O} \rightarrow \text{CH}_3\text{COOH}$ (acidified $\text{K}_2\text{Cr}_2\text{O}_7/\text{KMnO}_4$). [Q477, Q479, Q493, Q495]
- Addition reaction: Unsaturated compounds mein atoms add hote hain double/triple bond break hokar. $\text{C}_2\text{H}_4 + \text{H}_2 \rightarrow \text{C}_2\text{H}_6$ (hydrogenation). $\text{C}_2\text{H}_4 + \text{Br}_2 \rightarrow \text{C}_2\text{H}_4\text{Br}_2$ (bromine water decolourise). [Q479]
- Substitution reaction: Saturated compounds mein atom replace hota hai. $\text{CH}_4 + \text{Cl}_2 \rightarrow \text{CH}_3\text{Cl} + \text{HCl}$ (UV light). [Q477]
- Esterification: Carboxylic acid + Alcohol $\rightarrow$ Ester + Water (conc. H_2SO_4). $\text{CH}_3\text{COOH} + \text{C}_2\text{H}_5\text{OH} \rightarrow \text{CH}_3\text{COOC}_2\text{H}_5 + \text{H}_2\text{O}$; sweet smell. [Q503, Q505]
- Saponification: Ester + $\text{NaOH}/\text{KOH} \rightarrow$ Soap + Glycerol. [Q511]

8. Ethanol: Preparation, Properties & Uses

- Ethanol colourless liquid; pleasant smell; volatile; miscible with water. [Q481, Q483, Q485]
- Preparation (Fermentation): $\text{C}_6\text{H}_{12}\text{O}_6 \rightarrow 2\text{C}_2\text{H}_5\text{OH} + 2\text{CO}_2$ (yeast enzyme zymase, anaerobic). [Q481, Q483]
- Rectified spirit = 95% ethanol + 5% water; Absolute alcohol = 100% pure. [Q485]
- Denatured alcohol = ethanol + methanol/pyridine (non-potable). [Q485]
- Uses: alcoholic beverages, medicines, solvent, perfumes, gasohol (petrol + ethanol fuel). [Q487, Q489]
- Ethanol weak acid; sodium ke saath slow reaction: $2\text{C}_2\text{H}_5\text{OH} + 2\text{Na} \rightarrow 2\text{C}_2\text{H}_5\text{ONa} + \text{H}_2 \uparrow$. [Q489, Q491]
- Dehydration conc. H_2SO_4 se: $170^\circ\text{C} \rightarrow$ Ethene ($\text{C}_2\text{H}_5\text{OH} \rightarrow \text{C}_2\text{H}_4 + \text{H}_2\text{O}$). $140^\circ\text{C} \rightarrow$ Diethyl ether ($2\text{C}_2\text{H}_5\text{OH} \rightarrow \text{C}_2\text{H}_5\text{OC}_2\text{H}_5 + \text{H}_2\text{O}$). [Q495]

9. Ethanoic Acid: Properties, Reactions & Uses

- Pure ethanoic acid = glacial acetic acid (freezing point 17°C). [Q497, Q499]
- 5–8% solution = vinegar; food preservative, pickling. [Q499, Q501]
- Weak acid; organic acid hone ki wajah se partial dissociation. [Q501]
- Esterification: Alcohol ke saath sweet-smelling ester. [Q503, Q505]
- NaOH ke saath: $\text{CH}_3\text{COOH} + \text{NaOH} \rightarrow \text{CH}_3\text{COONa} + \text{H}_2\text{O}$. [Q507]
- Carbonates/bicarbonates ke saath CO_2 release: $2\text{CH}_3\text{COOH} + \text{Na}_2\text{CO}_3 \rightarrow 2\text{CH}_3\text{COONa} + \text{H}_2\text{O} + \text{CO}_2 \uparrow$. [Q507]
- Mg, Zn ke saath H_2 gas: $2\text{CH}_3\text{COOH} + \text{Mg} \rightarrow (\text{CH}_3\text{COO})_2\text{Mg} + \text{H}_2 \uparrow$. [Q509]

10. Soaps & Detergents

- Soap = sodium/potassium salts of long-chain fatty acids; saponification se banta hai. $\text{Fat/Oil} + \text{NaOH} \rightarrow \text{Soap} + \text{Glycerol}$. [Q511, Q513]
- Cleansing micelle formation se — hydrophobic tail grease mein, hydrophilic head water mein. [Q513, Q515]
- Soap hard water mein scum (Ca/Mg insoluble salts) banata hai; cleaning capacity kam. [Q515, Q517]
- Detergents = synthetic; petroleum se; $\text{Ca}^{2+}/\text{Mg}^{2+}$ se precipitate nahi; hard water mein better. [Q515, Q517]
- Soap biodegradable; kuch detergents non-biodegradable (water pollution). [Q517]
- Shampoo, washing powder = synthetic detergents. [Q517]

11. Petroleum, Natural Gas & Petrochemicals

- Petroleum = fossil fuel; crude oil; mixture of hydrocarbons. [Q489, Q491]
- Refining = fractional distillation; fractions alag boiling points par collect hote hain. $\text{LPG} < \text{Petrol/Gasoline} < \text{Kerosene} < \text{Diesel} < \text{Lubricating oil} < \text{Paraffin wax} < \text{Asphalt}$. [Q491]
- Natural gas mainly methane (CH_4); CNG = compressed natural gas. [Q493]
- Petrochemicals = petroleum se chemicals; plastics, synthetic fibres, rubber, detergents, medicines. [Q495]
- Methane (CH_4) = simplest hydrocarbon; marsh gas, gobar gas, CNG; greenhouse gas. [Q481]
- Aerated drinks (soda water) mein solute ke roop mein Carbon dioxide gas dissolved hoti hai. [Q488]

12. Plastics: Thermoplastic vs Thermosetting

- Thermoplastics = heat par soft, reshape ho sakte hain, recycle ho sakte hain. Examples: Polythene (PE), PVC, Polystyrene, Teflon. Uses: bags, bottles, toys, pipes. [Q497, Q499]
- Thermosetting plastics = ek baar set hone ke baad heat par melt nahi hote; permanently hard. Examples: Bakelite, Melamine, Polyester resin. Uses: electrical switches, handles, utensils, laminates. [Q499, Q501]
- ⚠ Thermoplastics recycle ho sakte hain; thermosetting nahi. [Q499, Q501]

13. Polymers: Natural vs Synthetic

- Polymers = macromolecules; repeating monomer units. [Q503, Q505]
- Natural polymers: Cotton (cellulose), Silk (protein), Wool (protein), Rubber (isoprene). [Q505, Q507]
- Synthetic polymers: Polythene, PVC, Nylon, Polyester, Teflon. [Q507, Q509]
- Monomers = small repeating units. Ethene $\rightarrow$ Polythene; Vinyl chloride $\rightarrow$ PVC; Isoprene $\rightarrow$ Rubber. [Q509, Q511]

- Nylon = polyamide; socks, ropes, parachutes.
- Polyester = terylene/cotton blend; fabrics, bottles. [Q511, Q513]
- Teflon = non-stick coating; cookware. [Q513]

14. Rubber: Natural & Synthetic

- Natural rubber = rubber tree ka latex; polymer of isoprene. [Q515, Q517]
- Natural rubber soft aur sticky; vulcanization se hard aur elastic. Sulphur ke saath heat karke cross-linking. [Q517]
- Synthetic rubber: Neoprene, Buna-S, Buna-N; tyres, hoses, belts. [Q517]

15. Environmental Impact & Non-biodegradable Substances

- Plastics non-biodegradable; soil aur water pollution. [Q517, Q518]
- Plastic burning se toxic gases (dioxins) release. [Q518]
- 4R principle: Reduce, Reuse, Recycle, Recover. [Q518]
- Biodegradable plastics research (starch-based). [Q518]
- CFCs ozone layer deplete karte hain; replaced by HFCs. [Q518]
- Ek carbon credit mein 1000 kg Carbon dioxide hoti hai (carbon trading/emission ka unit). [Q486]

16. ⚠ Traps / Key Differences Summary (Q448–Q518)

- ⚠ Trap: Carbon ke paas 2 shells hain; 4 valence electrons — tetravalent. [Q448]
- ⚠ Trap: Diamond hard + non-conductor; Graphite soft + conductor — allotropes hain. [Q451, Q453]
- ⚠ Trap: Fullerene (C₆₀) soccer ball shape; hexagons + pentagons. [Q455, Q478]
- ⚠ Trap: Alkane = single bond C_nH_{2n+2}; Alkene = double bond C_nH_{2n}; Alkyne = triple bond C_nH_{2n-2}. [Q454, Q456, Q458]
- ⚠ Trap: Cyclohexane mein 6 C-C + 12 C-H = 18 total covalent bonds. [Q449]
- ⚠ Trap: Homologous series mein successive members mein CH₂ difference; properties gradually change. [Q465, Q467]
- ⚠ Trap: Isomers ki molecular formula same; structural formula alag. [Q468, Q470]
- ⚠ Trap: Combustion: saturated = clean flame; unsaturated = sooty flame. [Q475]
- ⚠ Trap: Addition unsaturated mein; Substitution saturated mein. [Q477, Q479]
- ⚠ Trap: Fermentation mein yeast anaerobic condition mein kaam karti hai; oxygen ho toh CO₂ + H₂O banega. [Q481, Q483]
- ⚠ Trap: Glacial acetic acid = pure CH₃COOH; vinegar sirf 5–8% acetic acid. [Q497, Q499]
- ⚠ Trap: Esterification reversible hai; conc. H₂SO₄ catalyst + dehydrating agent hai. [Q503, Q505]
- ⚠ Trap: Soap mein -COO⁻Na⁺; detergent mein -SO₃⁻Na⁺ — functional group alag. [Q511, Q513]
- ⚠ Trap: Micelle tabhi form hota hai jab concentration CMC se upar ho. [Q513, Q515]
- ⚠ Trap: Thermoplastic = soft on heat, reshape, recyclable; Thermosetting = permanently hard. [Q499, Q501]
- ⚠ Trap: Vulcanization mein sulphur add karte hain cross-linking ke liye. [Q517]
- ⚠ Trap: Natural gas mainly methane (CH₄); LPG = butane/propane mixture — alag hain. [Q481, Q491]
- ⚠ Trap: Biodegradable = microorganisms se toot jaata hai; plastics non-biodegradable hain. [Q518]
- ⚠ Trap: Coal carbon ka allotrope nahi hai — Diamond/Graphite/Fullerene/Graphene hi allotropes hain. [Q500]

- ⚠ Trap: Carbon compounds mein bonding hamesha covalent hoti hai — ionic bond generally nahi milta. [Q466, Q476]
- ⚠ Trap: Diamond mein har carbon 4 dusre carbons se bonded hai; Graphite mein 3 (same plane mein) — structure difference. [Q502, Q504]